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Evaluation of Curcuma longa and Curcumin in Experimental Osteoarthritis and Synovial Inflammation

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ABSTRACT

Curcuma longa (turmeric), an ancient perennial herb belonging to the family Zingiberaceae, is extensively documented across traditional Ayurvedic medicine. The rhizome contains active bioactive polyphenols known as curcuminoids, predominantly curcumin. This investigation evaluates traditional ethnobotanical claims against modern in vitro assays, rodent models, and human clinical trial cohorts.

ETHNOPHARMACOLOGY AND TRADITIONAL USES

In Ayurvedic and Traditional Chinese Medicine, Curcuma longa has been traditionally used for centuries for the treatment of inflammatory disorders, joint rheumatism, biliary colic, and abdominal pain. Traditional preparations involved boiling the dried rhizome with milk or topically applying fresh rhizome paste onto septic lesions and sprains.

IN VITRO MECHANISTIC ASSAYS

In vitro macrophage cell culture assays revealed that curcumin inhibited the activation of NF-kappaB and downregulated COX-2 expression with an IC50 of 5.8 micromolar. Furthermore, curcumin exhibited marked free radical scavenging activity in DPPH antioxidant assays, neutralizing reactive oxygen species in synovial fibroblasts.

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IN VIVO ANIMAL MODELS
In male Wistar rats subjected to carrageenan-induced paw edema, oral administration of standardized Curcuma longa extract (250 mg/kg) produced a 62% reduction in paw swelling over 6 hours. Histopathological analysis confirmed significant suppression of neutrophil infiltration and preservation of articular cartilage integrity without noticeable gastric ulceration.

RANDOMIZED CONTROLLED CLINICAL TRIALS
In a randomized, double-blind, placebo-controlled clinical trial involving 120 patients with symptomatic knee osteoarthritis, oral supplementation with 1000 mg/day of standardized curcumin extract for 12 weeks demonstrated statistically significant pain reduction in Western Ontario and McMaster Universities Osteoarthritis Index (WOMAC) scores. Serum C-reactive protein (CRP) levels decreased by 34% compared to baseline in the active treatment cohort.

DISCUSSION AND CONCLUSIONS
These multi-tiered findings support the traditional ethnopharmacological status of Curcuma longa while identifying specific molecular targets of curcumin. Further multi-center human trials are recommended to validate long-term bioavailability.